ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 4TH YEAR

MECHATRONIC SYSTEM DESIGN

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BEng Mechatronics and Robotics Engineering, 4th year final examination: the engineering design process, requirements and concept selection, modelling a DC motor drive, actuator and battery sizing for a mobile robot, discrete-time (digital) control, reliability, FMEA and design for manufacture.

Written by Academic Central for practice, following final-year mechatronic design content at South African universities (e.g. University of Pretoria MOX 410 Design: ECSA GA3 engineering design, concept selection, analysis, manufacturing and costing; and MBB 410 Mechatronics: sensors, actuators and discrete-time control). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FIVE questions. Answer ALL the questions.
2. Show ALL your working, including formulae, substitutions and the assumptions you make. Give answers correct to TWO decimal places, with units.
3. Use g = 9.81 m/s².
4. Non-programmable scientific calculators may be used.
5. Number your answers exactly as the questions are numbered.

QUESTION 1: THE DESIGN PROCESS

1.1 List and briefly describe the main stages of the engineering design process, from the problem statement to the verified product. (5)

1.2 A company needs an autonomous robot to move crates in a warehouse. Explain the difference between a requirement and a specification, and write THREE measurable specifications for the robot. (5)

1.3 Three concepts for a robot gripper are scored out of 10 against weighted criteria. Use a weighted decision matrix to choose the best concept. (6)

Criterion	Weight	Concept A	Concept B	Concept C
Cost	0.3	7	9	6
Accuracy	0.4	8	6	9
Ease of maintenance	0.3	6	8	6

1.4 Explain the difference between verification and validation, and give an example of each for the warehouse robot. (4)

[20]

QUESTION 2: MODELLING A DC MOTOR DRIVE

A permanent-magnet DC motor has armature resistance R = 2 Ω, negligible inductance, torque constant K_t = 0.05 N·m/A, back-EMF constant K_e = 0.05 V·s/rad, rotor and load inertia J = 1 × 10^{-4} kg·m², and negligible friction.

2.1 Write the electrical and the mechanical equations of the motor. (4)

2.2 Derive the transfer function Ω(s)/V(s) and write it in the form K/(τs + 1). Give the values of K and τ. (6)

2.3 Calculate the no-load speed at 12 V, in rad/s and r/min. (2)

2.4 Calculate the stall torque at 12 V. (2)

2.5 Calculate the maximum mechanical output power at 12 V. (3)

2.6 How long does the motor take to reach 98% of its final speed after a step in voltage? (1)

2.7 How would including the armature inductance change the model? (2)

[20]

QUESTION 3: SIZING ACTUATORS AND THE BATTERY

A two-wheel-drive mobile robot has a total mass of 40 kg and wheels of radius 0.1 m. It must accelerate at 0.5 m/s² up a 5° slope and reach 1.2 m/s. The rolling-resistance coefficient is 0.02. Each driven wheel carries half the load.

3.1 Calculate the total driving force needed in the worst case (accelerating up the slope). (4)

3.2 Calculate the torque needed at each driven wheel. (2)

3.3 Calculate the wheel speed at 1.2 m/s, in rad/s and r/min. (2)

3.4 Calculate the peak mechanical power at each wheel. (2)

3.5 The motors have a rated speed of 3 000 r/min. Choose a gear ratio. (2)

3.6 The gearbox is 85% efficient. Calculate the motor torque needed. (3)

3.7 The robot uses an average of 20 W per motor plus 10 W for the electronics, and must run for 4 hours on a 24 V battery. Only 80% of the battery capacity may be used. Calculate the battery capacity in ampere-hours. (3)

3.8 Why does a designer add a safety factor when choosing the motors? (2)

[20]

QUESTION 4: DIGITAL CONTROL

4.1 Write the velocity (incremental) form of a discrete PI controller, and explain why it is useful on a microcontroller. (4)

4.2 The motor in QUESTION 2 has G(s) = 20 / (0.08s + 1). With a zero-order hold and a sampling time T = 0.01 s, the discrete model is G(z) = K(1 − a) / (z − a) with a = e^(−T/τ). Calculate a and write G(z). (5)

4.3 Write the difference equation of G(z), and calculate the first two output samples y[1] and y[2] for a unit step input that starts at k = 0 (y[0] = 0). (6)

4.4 Where must the poles of a discrete system lie for the system to be stable? (2)

4.5 Use the motor's rise time (about 2.2τ) to justify a suitable sampling time. (3)

4.6 What is integrator wind-up, and how is it prevented? (3)

4.7 Why does an encoder give a poor speed measurement at very low speeds, and how can this be improved? (2)

[25]

QUESTION 5: RELIABILITY AND MANUFACTURE

5.1 Explain what a failure mode and effects analysis (FMEA) is, and calculate the risk priority number for a failure with severity 8, occurrence 3 and detection 5. (4)

5.2 Reliability:

5.2.1 Three parts in series have reliabilities 0.95, 0.98 and 0.99. Calculate the system reliability. (2)

5.2.2 Two identical sensors, each with a reliability of 0.9, are used in parallel (either one is enough). Calculate the reliability of the pair. (2)

5.2.3 A controller has an MTBF of 2 000 hours and a constant failure rate. Calculate the probability that it runs 500 hours without failing. (2)

5.3 Give THREE good practices for a circuit board and wiring that combine motors and sensitive electronics. (3)

5.4 Explain design for manufacture and assembly (DFMA), with one example. (2)

[15]

TOTAL: 100

END OF QUESTION PAPER